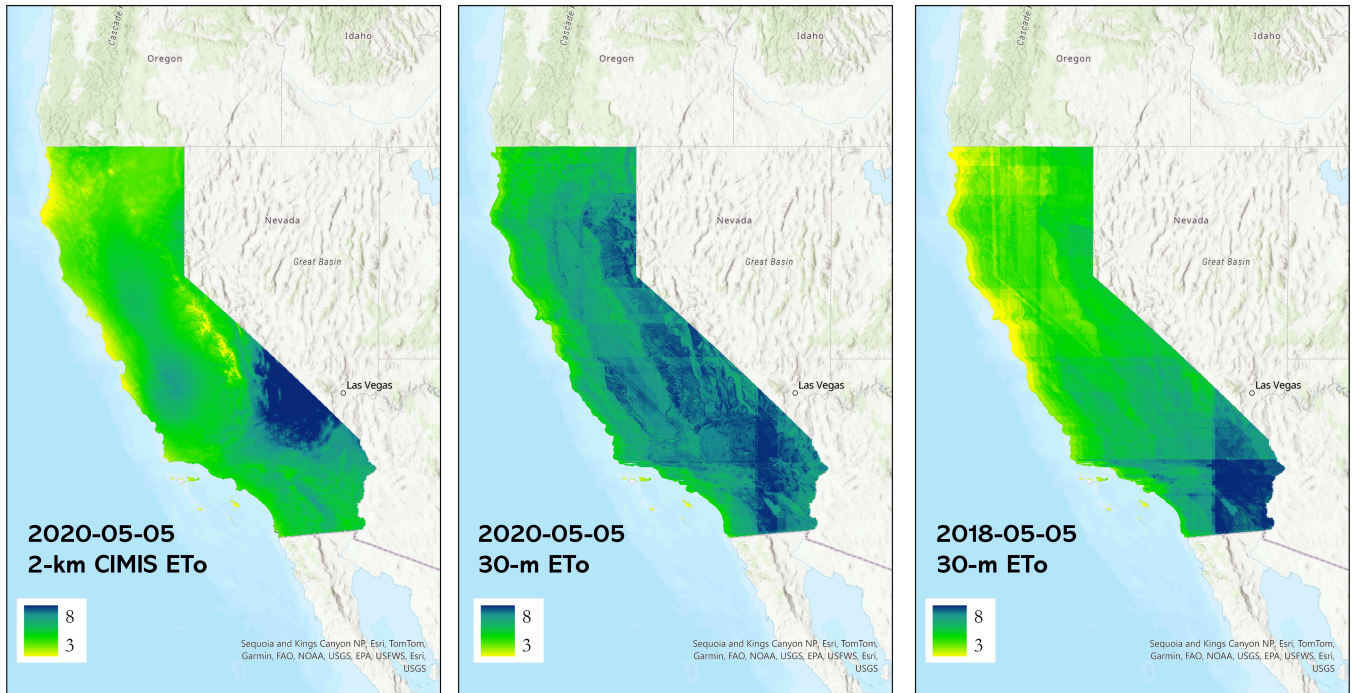




How do I get ETo from the spatial CIMIS and one 30-m ETo dataset?

*Once you open the CIMIS website (new version), you can quickly check data from the nearest station. If you want to know the ETo at a specific location for a specific time period, you may want to check this repository based on the **spatial CIMIS** dataset.*

Because we are working on high-resolution ETo, we also provide the corresponding ETo as a reference.

Requirements

1. A location within California state.
2. Raw spatial CIMIS data should be downloaded locally.
3. The 30-m ETo products should also be downloaded locally.

Features

After running this program, you are supposed to get:

1. The ETo extracted from the 2-km pixel resolution spatial CIMIS ETo maps at the location you defined.
2. The corresponding ETo extracted from another resource, 30-m pixel resolution spatial ETo maps, at the location you defined.
3. A time-series ETo comparison figure.
4. A *.CSV format document containing the ETo values and the corresponding time stamp.

Installation

```
pip install "git+https://github.com/RuiGao9/pyscimis.git"
```

How to Use This Repository for Data Downloading

The second section of `RunThis.ipynb` should be edited based on your questions. Like below:

```
# Do you want to save the output to a CSV file? If yes:
save_output = True # If no, change to False
# Fold location to save the final dataframe
output_path = r"C:\GitHub\pyscimis\output\ETo_Comparison.csv"

# Define the folder where the CIMIS data is stored
CIMIS_DATA_DIR = r"D:\1_Postdoc\0_Data\4_CIMIS_Spatial\Download"
# Define the folder where your 30-m ETo data is stored
# Detailed information about this dataset can ask Rui for more details
# Rui's Emails: Rui.Ray.Gao@Gmail.com or RuiGao@UCMerced.edu
MY_NC_DATA_DIR = r"E:\6_ETo_30m"
# Define the start and end dates
start_date = "2021-01-01"
end_date = "2021-12-31"
# Define the latitude and longitude for the point of interest
[latitude, longitude] = [37.354999, -120.414366]
```

Reference

Gao R., Safeeq M., Viers JH. A Python and Shell-based toolkit for spatial CIMIS data acquisition and geospatial preprocessing. Zenodo. 2026. <https://doi.org/10.5281/zenodo.19137639>

Gao, R., Safeeq, M., & Viers, JH. A geospatial toolkit for calculating distances to California's geographic boundaries (v0.1.0). Zenodo. 2026. <https://doi.org/10.5281/zenodo.19560836>

Gao, R., Safeeq, M., & Viers, JH. Topographic feature extraction via Python-based TauDEM algorithm (v0.1.0). Zenodo. 2026. <https://doi.org/10.5281/zenodo.19583013>

Gao R., Safeeq M, Viers JH. A Python Toolkit for Reference Evapotranspiration (ETo) Calculation Directly from Pandas DataFrames (Initial). Zenodo. 2026. <https://doi.org/10.5281/zenodo.19197914>

Gao, R., Safeeq, M., & Viers, JH. (2026). California 30-m daily reference evapotranspiration dataset (2020): Version 1.0. Zenodo. <https://doi.org/10.5281/zenodo.20388023>

Note: The 30-m ETo dataset is currently available for 2018–2025. At this time, only the 2020 data have been released online. To request data for other years, please contact Rui using one of the email addresses listed at the end of this document.

Citation

Gao, R., Safeeq, M., Viers, J.H. (2026). pyscimis: Automated time-series data extraction tool for spatial CIMIS dataset (Version Initial). Zenodo. <https://doi.org/10.5281/zenodo.20603074>

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Contact information if issues were found

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